

Phase 7 – Predictive & Prescriptive Analytics

Time Series Forecasting

Analyze historical shipment/order data to predict future trends using models like ARIMA or Power BI forecasting. Helps in planning operations and demand cycles.

Inventory Forecast

Predict future inventory requirements based on historical stock movement, lead time, and demand variability to avoid stockouts and overstock.

Demand Forecast

Estimate customer demand using historical sales data, seasonality, and trends to improve supply chain planning.

Route Optimization

Optimize delivery routes using distance, time, and cost factors. Helps reduce fuel cost and delivery time using algorithms or Power BI mapping visuals.

Customer Segmentation

Group customers based on behavior, purchase patterns, and value using clustering techniques like K-means.

Supplier Risk Prediction

Identify risky suppliers based on delay rate, defect rate, and cost variance using calculated risk scores.

Stock-out Prediction

Predict when inventory will run out using consumption rate and lead time analysis.

Overstock Prediction

Identify excess inventory based on slow-moving stock and low demand patterns.

Revenue Forecast

Predict future revenue using historical sales trends and growth rates.

ABC-XYZ Matrix

Classify inventory based on value (ABC) and demand variability (XYZ) to optimize inventory management.

EOQ Optimization

Determine optimal order quantity that minimizes total inventory cost (ordering + holding cost).

Safety Stock Optimization

Calculate buffer stock required to prevent stockouts considering demand variability and lead time.